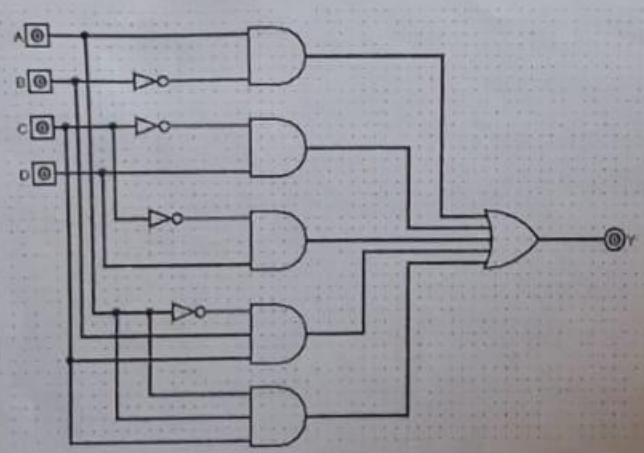
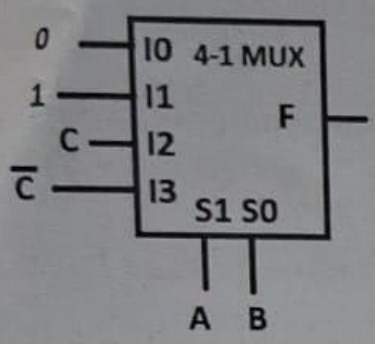
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Department of Computer Science and Engineering
CIE-I: Test Question Paper

Course Title **Course Code**	**Applied Digital Logic Design & Computer Organization (CS234AI)**	**Semester :**	**III semester**
Date : 12-10-2026	**Duration :** 90 Minutes	**Staff : KB/PSB/DD/NSK/DMN/KBR/NNN**	
Name:	**USN :**	**Section :**	A/B/C/D/E/F/CD/CY

1a	**Using Radix-4 Modified Booth's algorithm, multiply (−13) by 10. Show the following clearly:** a) Binary representation b) Formation of groups. c) Modified Booth encoding for each group. d) Generation of partial products, final product	5	L3	CO2
1b	Perform the division operation **18 ÷ 2** using the Restoring Division Algorithm. Show all the intermediate steps clearly and determine the final quotient and remainder	5	L3	CO1
2a	The combinational circuit shown in the figure is a function of four variables A, B,C, D a) Find its Boolean expression b) Minimize the Boolean expression found in (a) using Karnaugh Map c) Draw the resultant circuit 	5	L5	CO2
2b	Consider the decimal number **−18.625** a) Convert the number into binary b) Normalize the binary number. b) Determine the sign bit and biased exponent. c) Write the complete **32-bit IEEE754 single-precision**.	5	L2	CO1

3a	Realize the following Boolean function using a 4:1 multiplexer. Show the selection lines, determine the required multiplexer inputs, and draw the corresponding logic diagram. f (a, b, c, d) = ∑ m (1, 3, 4, 6, 10, 12,13,14)	5	L3	CO1
3b	Given the 4:1 MUX:  For the above circuit: a) Derive the Boolean expression for F b) Construct the truth table and identify the Sum of Products.	5	L5	CO2
4a	Design and explain a 4-bit Carry Look-Ahead Adder (CLA). Derive the carry equations using Generate and Propagate functions, and obtain the expressions for all carry outputs and sum bits. Draw the corresponding logic circuit.	5	L3	CO1
4b	Three sensors S2,S1,S0 generate a 3-bit status code. Design a circuit using 3 to 8 decoder such that: • Y0=1 when **exactly one** sensor is active. • Y1=1 when **exactly two** sensors are active. • Y2=1 when **all three** sensors are active. Y0,Y1 and Y2 are multiple outputs. Determine how the decoder outputs can be combined to realize the required functions.	5	L6	CO4
5	Minimize the given Boolean function using Quine-McCluskey tabular method and verify the obtained expression with K-map technique. f (a, b, c, d) = ∑ m (0, 5, 7, 8, 9, 10, 11, 14, 15) +d (1, 6)	10	L3	CO1

BT LEVELS	L1	L2	L3	L4	L5	L6	COS	CO1	CO2	CO3	CO4	CO5	CO6
MARKS	0	4	32	-	10	4		30	16	-	4	-	-